

High Sensitivity and High Speed SiPM Array Based Optical Wireless Receiver

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Abstract *1×2 SiPM receiver arrays are experimentally demonstrated in high-speed optical wireless links. 9.5 Gbps transmission is achieved using a 2× 10010 SiPM array, while the required irradiance at 8 Gbps is reduced by 6.25× compared to a single 10010 SiPM receiver. ©2026 The Author(s)*

Introduction

Optical wireless communication (OWC) is a promising technology to achieve the high capacity indoor communications required for 6G. OWC systems have a broad unlicensed band, with readily available high bandwidth sources which can be sourced from optical data communications [1]. OWC receivers, however, are more challenging due to the fundamental trade-off between detector area and bandwidth. Larger detectors have better sensitivity and field of view (FOV) but increasing detector area also increases capacitance restricting bandwidth. Avalanche photodiodes (APDs) are preferred in OWC links to enhance sensitivity due to their intrinsic gain, but they suffer from excess noise and have limited bandwidth for large active area detectors. Silicon Photomultipliers (SiPMs) are a promising alternative candidate for OWC receivers and can operate at a low irradiance levels offering larger area, high photon detection efficiency (PDE), high gain and moderate bandwidth, but they suffer from nonlinearity due to long recovery times.

Several OWC links with SiPMs have been demonstrated. 3.5 Gbps was achieved using on-off keying (OOK) and Decision Feedback Equalization (DFE) with a 9 mm² SiPM at 420 nm [2], 5 Gbps was demonstrated using OFDM [3]. Although SiPMs have highest PDE at 420 nm, they have also been tested at 850 nm where fast Vertical Cavity Surface Emitting Lasers (VCSELs) are readily available, in this case achieving 8.2 Gbps using OOK+DFE at an irradiance of 15.34 W/m² [1].

In this paper, we experimentally demonstrate 1×2 arrays of 1mm² and 9mm² SiPMs, showing a significant reduction in the required irradiance at higher data rates. The reduced irradiance requirement could be used to extend the transmission distance or to reduce the required gain of the light collection optics to extend the field of view and hence relax alignment tolerance. As a proof of concept, two-element Onsemi MicroFC 10010, and 30020 SiPM array receivers are demonstrated. Using an 850 nm VCSEL and 1×2 10010

SiPM array, 9.5 Gbps (OOK+ DFE) is achieved at an irradiance of 8.31 W/m² at a bit error rate (BER) threshold of 3.8×10^{-3} . At 8 Gbps the 10010 array also reduces the required irradiance by 6.25× compared to single 10010 SiPM (14.7 W/m² and 2.51 W/m² respectively).

SiPM Array Receiver

An SiPM is an array of Geiger mode single photon avalanche photodiodes (SPADs) connected in parallel. The combination of SPAD and quenching resistor is referred as a microcell. Commercial SiPMs are available with various microcell and detector sizes. However, for a fixed cell size, increasing detector area typically increases capacitance and slows the pulse response [4]. To prevent an increase in the RC time constant associated with a larger detector we propose an array of SiPMs with each SiPM having a separate electrical front end, increasing the effective detection area without increasing the capacitance seen by individual front-end circuits.

Each microcell operates at an over-voltage V_{ov} , and its recharge behaviour following an avalanche event can be described as:

$$V_{ov}(t) = V_{ov} \left(1 - \exp\left(-\frac{t}{\tau_{RC}}\right) \right), \quad (1)$$

where t is the time since the avalanche event and τ_{RC} is the recharge time constant. The probability of an avalanche event depending on the instantaneous over-voltage is given by [5]:

$PDE_{eff}(\lambda, t) = PDE_{full}(\lambda)(1 - e^{-V_{ov}(t)/V_{char}})$, (2) where V_{char} is a characteristic voltage. Since the avalanche charge is proportional to the instantaneous over-voltage, an insufficient recharge time results in a reduced effective avalanche charge ($Q_{eff} \leq Q_{full}$), leading to a high degree of nonlinearity.

For an array of N identical SiPMs, the total number of microcells is increased by a factor of N . At a given irradiance, this leads to a higher number of detected avalanche events, resulting in a larger output signal. To achieve a given data rate, the required number of detected events per bit is approximately the same for identical sensors under the same target BER. In an array, these events are distributed across a larger

number of microcells, reducing the average firing rate per microcell. This increases the average recharge time between successive avalanches, thereby increasing Q_{eff} and PDE_{eff} compared to a single SiPM, leading to signal enhancement beyond simple linear area scaling.

This improvement in effective signal amplitude directly improves the SNR at the receiver, which is critical for the DFE performance in post processing stage. DFE is strongly dependent on the signal SNR because decision errors propagate through the feedback path [5]. The improved SNR provided by the array reduces the probability of incorrect decisions and suppresses error propagation.

In practical terms, such an array configuration enhances receiver sensitivity super linearly and extends the achievable data rate without requiring higher irradiance, providing a scalable approach for high-speed SiPM-based OWC systems. The larger area of the detector also improves receiver FOV.

Data Transmission Experiment

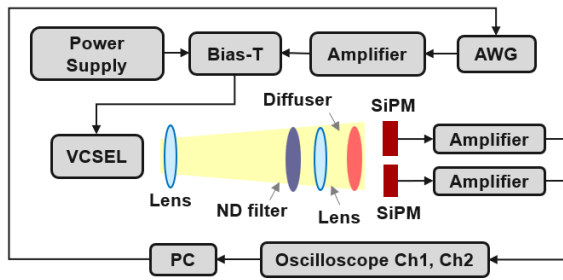


Fig. 1: Experiment Setup

To validate this concept, a 1×2 10010 array is implemented using two identical Onsemi MICROFC-SMA-10010 sensors, while 30020 array are realised on custom PCBs with Onsemi MICROFC-SMT devices. The experimental setup is shown in Fig. 1. Each SiPM output is connected to an independent electrical amplifier and scope channel. An 850 nm VCSEL biased at 1.82 V is used as the transmitter. An OOK data stream is generated by a Tektronix Arbitrary

Waveform Generator (AWG) and amplified (SHF 40 GHz amplifier) to 800 mV_{pp}. 8b10b coding is applied to minimize baseline wander effect from the SiPM. The VCSEL bias current is maintained at 7 mA for all data transmission experiments.

A spheric condenser lens (25.4 mm diameter) is placed in front of the VCSEL to collimate the beam. The SiPMs are positioned 25 cm from the transmitter. Sensor spacing is 0.7 mm for 10010 arrays due to the evaluation boards design and 0.25 mm for 30020 PCB array. A condenser lens and diffuser are positioned in front of the array to ensure uniform illumination with sufficient divergence to cover the whole SiPM array. The SiPMs are biased at the same voltage and are positioned to achieve the same photocurrent under uniform illumination. Irradiance is controlled by ND filters and measured with a power meter at the same location as the SiPM active area. The output of each SiPM is amplified separately and captured by two channels of a Tektronix MSO64 oscilloscope for offline processing using MATLAB. The received signals are low-pass filtered to remove out of band noise. A DFE with 80 feedforward and 20 feedback taps is applied to the combined signal to reduce inter-symbol interference (ISI). The maximum achievable data rate is determined at a BER threshold of 3.8×10^{-3} .

Experiment Results

The maximum achievable data rates at various irradiances for single and array receivers of different types of SiPMs: 10010 and 30020 are shown in Fig. 2a. The 10010 array achieved a higher data rate of 9.5 Gbps compared to 8 Gbps when single sensor was used. For the 30020, the data rate increased from 4.3 to 5 Gbps using the array. Large sensors (30020) perform better at low irradiance while small sensors (10010) achieve higher data rates at high irradiance. This highlights a trade-off between sensitivity and bandwidth.

The arrays consistently achieve higher data rates than single sensors at the same irradiance,

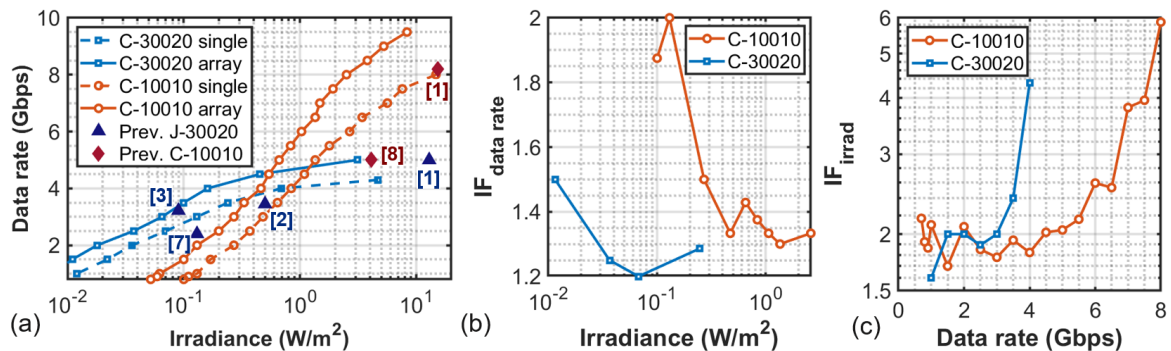


Fig. 1: (a) Achievable data rate vs irradiance using different receivers with DFE (b) Data rate improvement factor vs irradiance (c) Irradiance improvement factor vs data rate.

or equivalently require lower irradiance to reach a given data rate, as quantified in Fig. 2b and c. IF_{data_rate} gives the relative improvement in data-rate over the results at the same irradiance, $IF_{irradiance}$ gives the irradiance reduction factor at the same data rate. In Fig. 2b, the data rate improvement factor decreases with increasing irradiance for both sensor types. The improvement is highest at low irradiance especially for small sensors (up to 2x for 10010), where performance is limited by the SNR. In this regime, the single sensor is SNR limited and has excess available bandwidth, using the array the SNR improvement is allows further bandwidth to be exploited. At high irradiance, the improvement become constrained ($\sim 1.3x$ for both types) because the system is bandwidth limited. In this case, the SNR gain provided by the array allows only a small increase in the equalizer compensated bandwidth.

Fig 2c shows that the improvement in required irradiance increases rapidly with data rate due to SiPM nonlinearity. At higher data rates, a larger photocurrent is needed to maintain SNR, pushing the avalanche rate toward the microcell recovery limit. In this regime, the photocurrent compresses, so small current increases require disproportionately large increases in irradiance, making single devices inefficient. In contrast, arrays operate at a lower per-cell firing rate for the same total count, keeping microcells further from saturation and allowing arrays to remain in a region where photocurrent scales more efficiently with irradiance.

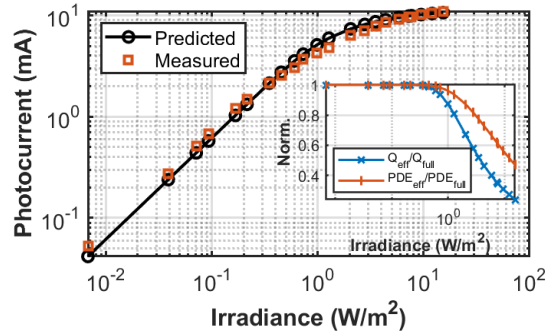


Fig. 3: C-series 10010 SiPM photocurrent as a function of irradiance. Inset: Normalized effective avalanche charge and PDE vs irradiance.

Fig. 3 shows the photocurrent as a function of irradiance for the 10010 SiPM considering the effect of different recharge times on Q_{eff} and PDE_{eff} . At low irradiance ($\leq 1 \text{ W/m}^2$), the photocurrent increases linearly, indicating sufficient active microcells. As the irradiance increases, the response starts to compress. This transition is further explained by the inset where both estimated Q_{eff} and PDE_{eff} drop with increasing irradiance due to insufficient recharge time between successive avalanches, reducing photocurrent generation efficiency. For a given total count rate, an

array distributes avalanches across more microcells, reducing the firing rate per cell and allowing longer recharge time, maintaining a higher Q_{eff} and PDE_{eff} . Therefore, arrays require less irradiance to achieve the same signal level and support higher data rates at a given optical power.

Comparison with previous work

Results from previous work are plotted in Fig. 2a for comparison. The operating conditions of each work are summarised in Table 1. Prior work used J-series 30020 while this work uses the C-series with slight higher (0.1 ns) fast out pulse FWHM. It should be noted that the PDE is higher at 405 nm, as this wavelength is close to the peak at 420 nm where the PDE is 18% for 10010 and 48% for 30020. The PDE at 850 nm drops to less than 1% for 10010 and 2.5% for 30020. This work uses a wavelength of 850 nm due to the availability of high-speed VCSELs, higher eye safety limit and ability to filter ambient light at this wavelength. Despite higher PDE at 405 nm, the presented array still outperform all previous results in terms of irradiance and data rate confirming the advantage of IR wavelengths in high speed OWC systems.

Tab. 1: A Summary of operating conditions in prior SiPM-based OWC systems.

Ref	SiPM	Wave-length	Equalization
[7]	J-30020	405 nm	DFE
[2]	J-30020	405 nm	DFE
[3]	J-30020	405 nm	Volterra
[1]	J-30020	850 nm	DFE
[8]	C-10010	405 nm	DFE
[1]	C-10010	850 nm	DFE

Conclusion

A 1×2 SiPM array-based receiver is demonstrated using 1 mm^2 and 9 mm^2 SiPMs, doubling the detector area in an OWC link with no reduction in 3 dB bandwidth. This improves sensitivity compared to a single SiPM at a constant irradiance, or enables a higher data rate at a fixed irradiance. With an 850 nm VCSEL source and OOK+DFE, the 10010 SiPM array achieved 9.5 Gbps under 8.31 W/m^2 and provide a 6.25x reduction in required irradiance (from 14.7 W/m^2 to 2.51 W/m^2) at 8 Gbps compared to single SiPM sensor. At high data rates, the improvement factor in the irradiance level increases due to single SiPM's heavy saturation of photocurrent response. Integration of the SiPM and electrical amplifiers could allow larger arrays, thus enabling high data rate OW to operate without receiver lenses for ultra wide field of view.

Acknowledgements

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